

任邵康

电话: (+86) 13983247522 | 邮箱: rsk23@mails.tsinghua.edu.cn

学校地址: 北京市海淀区清华大学紫荆学生公寓 2 号楼 邮编: 100084

家庭地址: 重庆市沙坪坝区南开中学柏树村 23 幢 3 单元 1 楼 2 号 邮编: 400030

微信号: Eroica_R

教育背景

工学学士 (计算机科学与技术), 清华大学, 中国, 2027

- GPA: 3.70 / 4.0

技术能力

编程语言与工具

- C / C++、Python、Java、JavaScript、汇编语言、SystemVerilog
- 熟悉 Linux 环境
- 熟悉机器学习与深度学习基础

数学基础

- 微积分
- 线性代数
- 离散数学
- 数据结构与算法
- 复变函数
- 形式语言与自动机

其他相关方向

- 计算神经科学
- 神经环路建模
- 信号处理
- 漏积分-发放 (Leaky Integrate-and-Fire, LIF) 神经元模型

- 泊松脉冲序列 (Poisson Spike Trains)

科研经历

果蝇昼夜节律神经环路的仿真与建模

2024 年 7 月 – 至今

导师：胡晓林 教授

清华大学 脑与智能实验室

- 旨在构建可信的果蝇昼夜节律神经环路计算模型，以揭示关键神经元及整体环路同步机制的动力学特性。
- 系统调研了神经环路计算建模的相关文献，重点关注果蝇昼夜节律的仿真模型。
- 深入理解了漏积分-发放模型 (LIF) 及其在神经动力学模拟中的应用。
- 将不同细胞中“快速钙信号”和“慢速钙信号”的昼夜变化模式映射至电导参数与放电阈值，基于果蝇连接组 (connectome) 与 Brian2 框架构建昼夜节律神经环路模型。
- 调研了泊松脉冲序列建模方法，以及 V1-V4 视觉皮层在视觉处理中的作用，并探讨其在昆虫神经系统中的潜在对应关系。

面向计算机与无线网络的 LLM 应用研究

2025 年 1 月 – 2025 年 2 月

导师：Yuchen Liu 教授

北卡罗来纳州立大学 计算机科学系

- 研究目标为在多种环境下对无线信号特征进行建模与预测，完成视距/非视距 (LoS / NLoS) 分类、室内定位等任务。
- 使用 PyCharm 对大规模无线模型 LWM 进行微调与应用，生成高质量嵌入表示，探索大语言模型在下游无线任务中的能力。
- 掌握了多头注意力机制、MIMO 数据集的使用方法，以及带残差连接的层归一化等基础概念。

课程项目经历

七级流水线设计

2025 年秋季

- 使用 SystemVerilog 设计了完整的七级流水线

在线实时聊天软件后端	2025 年春季
使用 Typescript 实现简易版类微信在线聊天软件后端	
内存分配器 (Malloc Lab)	2024 年秋季
使用 C++ 与汇编语言实现了一个内存分配器。	
数据结构与算法课程项目	2024 年秋季
- 使用 C++ 复现多种经典算法与数据结构，包括： - 高级平衡二叉搜索树 - 图算法 - 哈希结构 - 字符串匹配算法	
Java 新闻客户端	2024 年夏季
- 实现新闻抓取与展示、本地缓存等基础功能。 - 集成 GPT 实现新闻摘要自动生成。	

荣誉奖项

- 清华大学新生奖学金 (2023-至今)
- 清华大学计算机系文艺奖学金 (2023-至今)

课余经历

- 清华大学交响乐团 (TUSO) 二队小提琴首席兼财务负责人
- 清华大学计算机系学生节话剧《罗生门》导演兼编剧兼主演
- 清华大学“计舟桂行”赴广西实践队宣传负责人
- 托福英语考试 109, GRE 研究生考试 329, 擅长意大利语, 熟悉日语
- 清华大学计算机系计 36 班文艺委员

Shaokang Ren

Tel: (+86)13983247522 | Email: rsk23@mails.tsinghua.edu.cn

EDUCATION

Tsinghua University, Department of Computer Science and Technology
Beijing, China

Bachelor of Engineering in Computer Science and Technology
Sep 2023–Jul 2027(expected)

- Overall GPA: 3.70/4.0

- **Technical Skills:**

- **Programming Skills:** C/C++, Python, Java, Java Script, Assembly Language. Familiar with Machine Learning and Deep Learning, Familiar with Linux.

- **Mathematics:** Calculus, Linear Algebra, Discrete Mathematics, Data Structure and Algorithms, Functions of Complex Variables, Formal Language and Automata

- **Other:** Computational Neuroscience, Neural Circuit Modeling, Signal Processing, Leaky Integrate-and-Fire Model, Poisson Spike Trains

RESEARCH EXPERIENCES

Simulation and modeling of circadian neural circuits in *Drosophila melanogaster*

Jul 2024 –present

Supervisor: Prof. Xiaolin Hu, Tsinghua Laboratory of Brain and Intelligence, Tsinghua University

- Aimed to provide a convincing simulation of the circadian neural circuits in *Drosophila melanogaster*, thus uncovering the dynamics and function of the circuits like the key neurons and synchronization of the whole circuits.
- Conducted an extensive literature review on computational models of neural circuits, particularly focusing on the simulation of circadian rhythms in *Drosophila melanogaster*.
- Gained a comprehensive understanding of the leaky integrate-and-fire model and its applications in simulating neural dynamics.
- Mapping the distinct daily patterns of fast calcium and slow calcium activity of different cells to conductance and voltage threshold, thus simulating the circadian neural circuits in *Drosophila melanogaster* based on the *Drosophila* connectome and brian2 frame.
- Reviewed advancements in modeling Poisson spike trains and the role of V1-V4 brain structures in visual processing, including potential parallels in insect models.

LLM for computer and wireless networking

Jan 2025 –Feb 2025

Supervisor: Prof. Yuchen Liu, Department of Computer Science, North Carolina State University

- Aimed to process and predict the feature of wireless signals in various environments, completing multiple tasks such as LoS/NLoS classification and indoor localization.
- Leveraging and fine-tuning LLM, a large wireless model using pytorch to generate rich embeddings, gaining practical insights into the capabilities of LLM for downstream tasks.
- Grasped an understanding of fundamental concepts like multi-head attention, the use of MIMO datasets and layer normalization with residual connection.

SELECTED COURSE PROJECTS

Seven-Stage Pipeline Processor Design

Fall 2025

Designed and implemented a complete seven-stage pipeline using SystemVerilog

Online Real-Time Chat Application Backend

Spring 2025

Implemented a simplified WeChat-like online chat backend using TypeScript

Malloc lab

Fall 2024

- Achieved a memory allocator using C++ and Assembly Language.

Data Structures and Algorithms Projects

Fall 2024

- Reproduced classic algorithms based on C++ (Advanced binary search trees/Graph/Hash/String match).

Java-based News client

Summer 2024

- Realized basic grasping and presenting of daily news, local caching, and other functions.
- Realized abstract generating by incorporating GPT.

HONORS

- Tsinghua University Freshman Scholarship
2023 -present
- Tsinghua University Art Scholarship of Computer Science and Technology Department
2023 -present

EXTRACURRICULAR ACTIVITIES

- Concertmaster (Principal Violin) and Financial Officer of the Second Orchestra, Tsinghua University Symphony Orchestra (TUSO)
- Director, Playwright, and Lead Actor of the Tsinghua University Department of Computer Science student drama production Rashomon
- Publicity Lead of the "Jizhou Guixing" Social Practice Team to Guangxi, Tsinghua University
- TOEFL: 109; GRE: 329; Proficient in Italian, with working knowledge of Japanese
- Arts and Culture Representative, Class CS36, Department of Computer Science, Tsinghua University